

DIAGNOSIS BUYS A SHOPPING LIST, NOT A RATIO: BENCHMARK-GUIDED DATA REPAIR AND ITS GENRE- GATED RETURNS

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ABSTRACT

[TODO: abstract written last per C-12; thesis sentence source = CLAIMS CL-5: (i) genre determines the visibility of repair-data gains (+10–16pp in the repair genre vs +2–7pp on plain items); (ii) profile-matched ratios show no advantage in either genre — the actionable output of diagnosis is a shopping list (which components must be present, which are toxic, in which genre to validate), not a mixing ratio.]

1 INTRODUCTION

[TODO: written last (C-12). Spine: Fig. 1 invoked in paragraph 1; thesis = CL-5 two sentences; named alternative rejected = “diagnosis buys a mixing ratio”.]

2 DIAGNOSTIC AND REPAIR PROTOCOL

Our pipeline has three stages: a perturbation-probe suite that turns a benchmark score into a failure profile (§2.1), a library of repair-data components and mixing arms trained from that profile (§2.2), and a two-genre evaluation protocol with a fixed operating-point rule that separates repair gains from collateral damage (§2.3). Throughout, the subject is a single pre-repair model (Qwen3-8B instruct); all decoding is greedy with thinking mode disabled.

2.1 PROBE SUITE: FROM ONE SCORE TO A FAILURE PROFILE

For each GSM8K test item we generate a family of eight probes that hold the underlying arithmetic task fixed and perturb one interface at a time (Table 1). The original item (**O**) anchors the family. Three probes supply *assistance*: a step scaffold in verb-phrase form (**S**), an entity-free statement of the governing rule (**R**), and a correct intermediate value (**C**). Two probes plant *misinformation* in the context: a wrong intermediate value (**W1**) and a wrong final claim (**W2**); both record the planted value so that answering with it (*adopt*) can be distinguished from answering with any other wrong value (*derail*) or not answering (*mute*). One probe rephrases the surface text (**P**), and one requests the identical computation under a JSON output constraint (**F**). Assistance and misinformation probes are validated by leakage filters (scaffolds and rules must contain no digits and no problem entities); every probe type passed a manual construct audit of 100 items per type with a preregistered $\geq 70\%$ acceptance gate. The pool is GSM8K test (1,300 items) plus a filtered GSM-hard bucket; 330 items with negative or non-integer golds are excluded from probe construction.

A frozen classifier maps each item’s eight-bit signature to a failure bucket: *conduct* (solves O but is flipped by W1/W2), *format* (only F differs), *phrasing* (O and P disagree), *scaffold / rule* (fails O, recovered by S / by R alone), *local execution* (recovered only by C), and *unresolved* (fails everything). Mixed signatures go to an explicit mixed bucket rather than being forced into a single label. The resulting profile is the paper’s diagnostic object (Figure 1).

Probe	Perturbs	Holds fixed	Failure it isolates
O	— (original)	everything	baseline correctness
P	surface wording	task, numbers	phrasing sensitivity
S	+ step scaffold	task	missing-plan (scaffold) failures
R	+ explicit rule	task	missing-rule failures
C	+ correct intermediate	task	local execution failures
W1	+ wrong intermediate	task	credulity (conduct)
W2	+ wrong final claim	task	credulity (conduct)
F	output format (JSON)	task, wording	format-coupled computation

Table 1: The probe suite. Each GSM8K item yields one probe family; a frozen rule set maps each family’s pass/fail signature to a failure bucket.

2.2 REPAIR COMPONENTS AND MIXING ARMS

Each failure bucket names a candidate repair-data *component*; the components are deliberately minimal so that mixtures can be credited to their parts. **Conduct**: worked examples that verify a planted intermediate, reject it on mismatch, and recompute. **Format**: the same computation performed correctly *under* the JSON constraint. **Phrasing, scaffold, rule**: LLM-generated pools (kept 2,200 / 1,606 / 1,125 after validator gates with preregistered minimum-yield thresholds). **Drills**: bare “compute 17×23 ” items, included as a preregistered falsification component. **Replay**: clean GSM traces.

Arms are composed from these pools under a preregistered design (single-component arms: 600 items each, fixed; mixture arms: 2,000 items, character-aligned within 2%): **A1** mixes components in proportion to the measured failure prevalences (40/36/16/5/2 for conduct/format/phrasing/scaffold/rule); **B** mixes them uniformly; **D** applies the frozen reverse permutation of A1 (2/5/16/36/40); **C** is generic chain-of-thought data with no repair design; and **clean replay** (600 items) serves as the *training placebo*, isolating the effect of any fine-tuning at all. Dose arms replace N items of the fixed replay carrier with one component (format at 10/20/36%; drills at 10/25%), so the dose axis varies only the component share. All training runs use identical hyperparameters (full fine-tuning, $\text{lr } 10^{-5}$ cosine, epochs $\in \{2, 4, 8, 16\}$; A1, B and C are run with three seeds).

2.3 TWO-GENRE EVALUATION AND THE OPERATING-POINT RULE

Every checkpoint is evaluated in two genres. The plain genre re-runs the full probe suite: rescue is counted per bucket, in two frozen readings — the *strict* reading requires an item’s whole signature to turn clean; the *loose* reading requires only the bucket’s defining probes to flip from fail to pass — and every rescue number is reported as a delta against the clean-replay placebo, because placebo rescue is large in some buckets (§4). The repair genre presents 480 corrupted-context items (planted wrong steps, wrong claims, clean controls, and unanswerable cases) under a JSON decision contract; scoring reuses the frozen strict-abstain judge of the accounting study, so numbers are comparable across chapters.

Checkpoints are compared only at their *operating point*: the smallest epoch satisfying three health gates on plain items — answer rate ≥ 0.95 , JSON bleed-through $\leq 5\%$, mute rate $\leq 12\%$. The gate is applied per arm; when no epoch passes, the arm is reported as having *no clean operating point* rather than being scored at its best epoch. This rule is what separates a repair gain from a checkpoint that has merely traded general usability for the target metric.

3 THE DIAGNOSTIC PROFILE: WHAT A BENCHMARK SCORE HIDES

[TODO: results prose after figure package approval. Numbers: surface 93.5 vs robust 60.6; buckets conduct 16.4 / format 14.7 / phrasing 6.4 / ability 3 / unresolved 1.9; flip-overlap core/fringe footnote (conduct core 57% placebo-fixable). CLAIMS: CL-4, profile v1.]

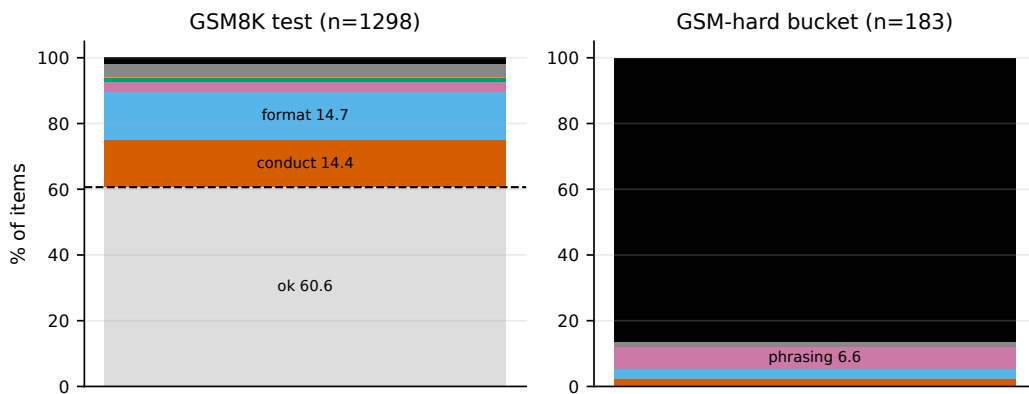


Figure 1: [TODO: three-piece set from figs/out/fig1-profile meta; footnote trio (denominator / pool filter / labile discount) per ruling.]

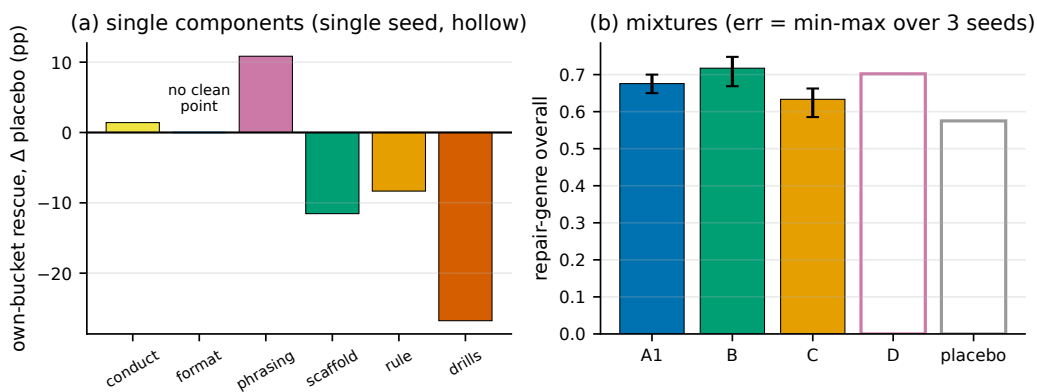


Figure 2: [TODO: panel (a) six single components; (b) mixtures vs placebo; hollow = single seed. CLAIMS: CL-5, COR-1.]

4 FROM PROFILE TO RECIPE: COMPONENTS MATTER, RATIOS DO NOT

[TODO: prose after figure approval. Safe sentence only (ruling: B>A1 red card): “matched never exceeded uniform in any reading.”]

5 GENRE GATING: THE GENRE IS THE SWITCH

[TODO: prose after figure approval; three independent measurements, scaffmt attribution known-limitation.]

6 MECHANISM AND BOUNDARY

[TODO: prose after figure approval; drills ΔW off-manifold (COR-3): decorrelated, not anti-aligned.]

7 THE PRESCRIPTION MANUAL

Diagnosis pays for itself not by tuning mixture ratios but through five rules, each backed by a dose curve or a cross-genre reconciliation:

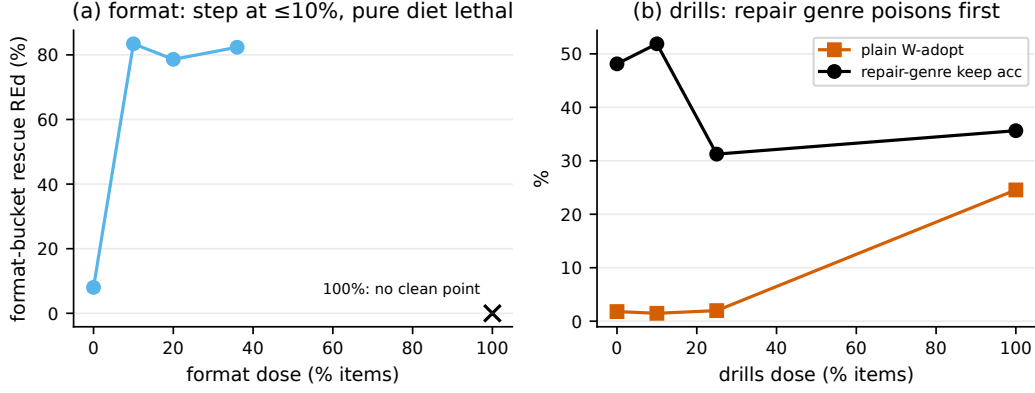


Figure 3: [TODO: format saturation step at $\leq 10\%$ dose; drills threshold + genre-gated harm; 100% points marked as no-clean-point / catastrophe. CLAIMS: CL-5, COR-2.]

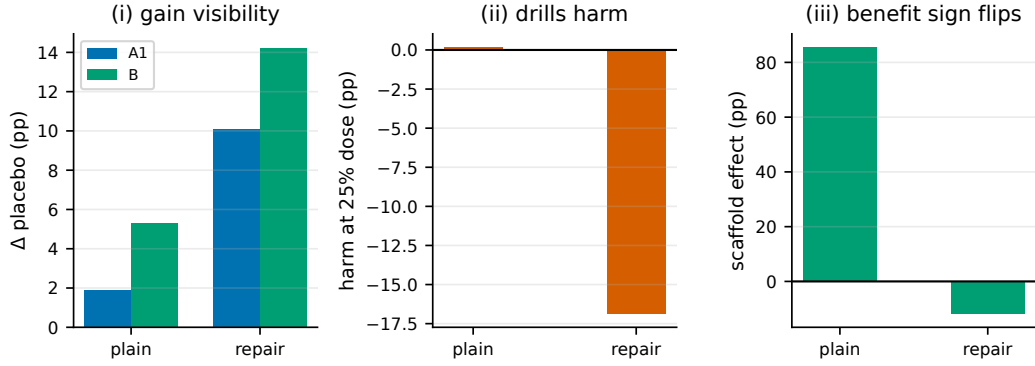


Figure 4: [TODO: triptych: gain visibility / drills harm / scaffold sign flip; same visual language. This figure IS the named claim. CLAIMS: CL-6.]

1. **Add the specific component; 60 items suffice.** Format rescue saturates at a 10% dose (a step, not a slope); further dose buys nothing.
2. **Test toxic components in the repair genre.** Compliance-style data (drills) is harmless on plain items up to a 25% dose yet already poisons repair-genre behaviour at 25%; plain-genre clearance is not clearance.
3. **The pure diet is the only lethal form.** 100% format data leaves no clean operating point (mute-side collapse); 100% drills teaches adoption of planted values (W_{adopt} 25%).
4. **Ratios do not matter; uniform coverage suffices.** Matched ratios never exceeded uniform in any reading.
5. **Deep credulity routes to targeted repair-genre data or steering; ability failures route to tools.** The plain-genre adopt subset ($n=31$) is rescued by no data arm at this budget.

Every step of the manual states the genre in which it is validated (§5).

8 RELATED WORK

[TODO: differential lines incl. natural-set sources (one line each, neighbour-not-first discipline: GSM-IC $\leftrightarrow$ W1, format-constraint literature $\leftrightarrow$ CL-4).]

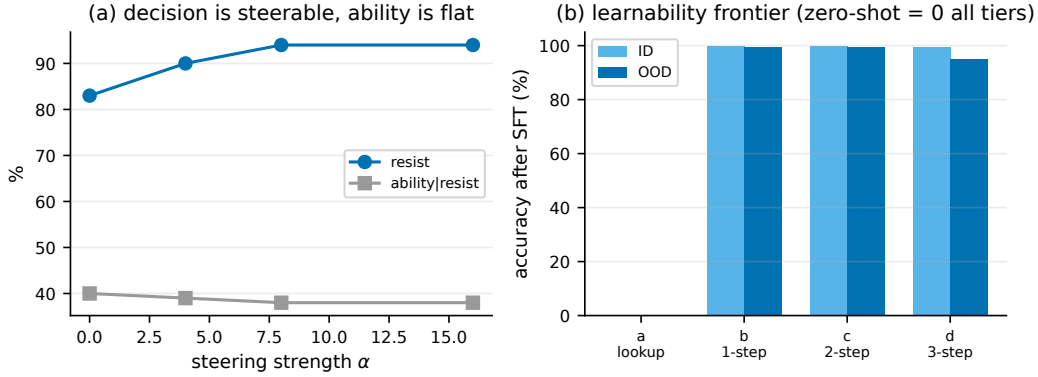


Figure 5: [TODO: left: steering four curves (83→94, ability flat); right: learnability frontier tiers a–d (zero/ID/OOD). CLAIMS: CL-2, CL-3.]

9 CONCLUSION

[TODO: written last; restates CL-5 + CL-6.]

REFERENCES

A PREREGISTRATION SCORECARD

Every training experiment in this paper was preregistered before any run started; this table reports every prediction against its outcome, including the misses — the misses are where the paper’s replacement hypotheses come from.

Prereg	Prediction	Outcome	Verdict
Loop 3 (recipe)	conduct coefficient .95 rule coefficient .90 format coefficient .60 phrasing / scaffold .50± drills ≈ 0 A2 arm (measured ratios)	specific effect +2pp over placebo −9pp ($n=12$) no clean point (precondition failed) 51% / 42% raw direction right, harm unpredicted inputs falsified in 4/5 buckets	MISS MISS MISS HIT / HIT (raw) dir. HIT non-executable
Batch 2 (dose)	all format doses pass gates; saturate by 20% drills harm monotone (≥ 5 pp at 10%) scaffmt separates structure vs JSON	all pass; saturates at $\leq 10\%$ flat on plain items; genre-gated falls in JSON interval (attribution stayed)	HIT* MISS HIT (rule)
Loop 5 (ΔW)	$\cos(d, \Delta W_{\text{drills}}) < 0$ single-component ΔW angles	+0.13: decorrelated, not opposed drills = global outlier (all pairwise min)	MISS registered
Tier 2 (frontier)	lookup tier A ≈ 0 ; depth decay M calibration formula	0.2/0.0%; monotone but no bend ≤ 3 steps self-neutralised at ceiling (no range)	5/5 HIT instrument note
E1 (data size)	targeted $N \leq 300$ suffices; random unstable	mute-wave artifact; measure replaced	hard stop

Table 2: Preregistration scorecard. *the named falsifier “already saturated at 10%” also fired — the effect is cheaper than predicted; we report both. The Loop-3 misses motivated the placebo-delta reading (COR-1); the dose miss became the genre-gating result (§5); the ΔW miss reframed “reverse vaccine” as behavioural-only (COR-3).

Abstain variance warning. The retrieve-or-abstain policy is the highest seed-variance column in the study (uniform arm across three seeds: .34/.15/.45); a single-seed ordering anomaly ($D > B > A1$) collapsed entirely into this band and is recorded as noise, not finding.

B INSTRUMENTS AND AUDITS

[TODO: abstain seed-variance warning table; flip-overlap matrix; construct audit protocol; judge definitions (strict/loose readings, non-dict guard).]